

Reasoning & Proofs

US Common Core – High School Geometry

Domain: Congruence (G-CO) | Cluster: Understand congruence in terms of rigid motions; make geometric constructions

1 Conditional Statements

Definition

A conditional statement has the form “If *hypothesis*, then *conclusion*.” The part following **If** is the **hypothesis**; the part following **then** is the **conclusion**.

If **the sun is scorching**, then **the ice cream will melt down**.

Hypothesis the sun is scorching

Conclusion the ice cream will melt down

2 Related Conditional Statements

Definitions

Given a conditional statement “If p , then q ”:

- **Inverse**: If *not* p , then *not* q .
- **Converse**: If q , then p .
- **Contrapositive**: If *not* q , then *not* p .

Original If the sun is scorching, then the ice cream will melt down.

Inverse If the sun is *not* scorching, then the ice cream would *not* melt down.

Converse If the ice cream is melting down, then the sun was scorching.

Contrapositive If the ice cream is *not* melting down, then the sun was *not* scorching.

Note: The contrapositive of a true conditional statement is always true; the converse and inverse are not guaranteed to be true even when the original statement is.